

A Systematic Review for Evaluating IoT Security: A Focus on Authentication, Protocols and Enabling Technologies

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Abstract—The idea of the Internet of Things (IoT) is to connect objects and smart devices with any digital element seamlessly at anytime, anywhere, with anything. These devices or objects are characterized by low power consumption, lower bandwidth usage, limited processing power, and limited memory capacity. This article presents a systematic literature review (SLR) of the state-of-the-art security challenges of IoT systems designed for authentication. It reports the strengths, weaknesses, threats, and attacks associated with the proposed authentication models. In addition, a comprehensive analysis of IoT-compatible protocols, enabling technologies, and countermeasures to mitigate risk in IoT is investigated. This study uses the PRISMA methodology to review peer-reviewed articles published from 2014 to 2023, studying 100+ papers by selecting approximately 10 papers each year. It provides a detailed overview that contrasts and compares the advances in securing IoT devices to date while identifying remaining open research questions for IoT security designs. This article highlights the need for a survey integrating authentication models with compatible protocols and other lightweight technologies. The examination and analysis in this work contribute to the body of knowledge and serve as a roadmap for researchers and practitioners to improve their work while designing IoT security models.

Index Terms—Attacks, authentication, Internet of Things (IoT), protocols, RFID, security, technologies.

I. INTRODUCTION

IOT (Internet of Things) provides seamless connectivity among a pool of networked objects, smart devices, and any physical entity with a digital element [1]. A definition by Atzori et al. [2] That: ‘IoT is a dynamic network topology with self-configuring abilities based on a set of interoperable protocols where the physical and virtual items have labelled as their identities, physical features, and virtual attributes’. IoT, in other words, can be seen as a network of objects in

which everything is identified using Internet Protocol version 6 (IPv6), connected via some technology, such as RFID, Bluetooth, ZigBee, Wi-Fi, etc., [3], and using computing devices, such as sensors, actuators, phones, cameras, laptops, PDAs, so that the objects interact and communicate to provide integrated services [4].

IoT is a new wave in Information Technology (IT) that is penetrating several domains, such as healthcare, transportation, marketing, military, surveillance, security, logistics, banking, e-commerce, smart homes, and many others [5], [6]. The concept of IoT is to provide connectivity everywhere and anytime in any network so that any service can be provided for everyone. IoT paradigms are becoming more and more popular every day. It connects all smart devices around us both locally and globally via Wireless Local Area Network (WLAN) and Internet wireless personal area network (WPAN) [7]. Many IoT applications are being developed daily and are changing how we live [8].

Kevin Ashton is reported to be the first to coin the term “Internet of Things” in 1991. The massive connectivity of smart IoT devices distributed physically broadens computation and communication to objects of various specifications. These devices’ sensing capabilities allow them to collect real-time data directly or remotely from the real world, share it, and send it for processing accordingly. IoTs are mounted on networked sensors or actuator systems, and the end devices of IoTs can collect information and make data-based decisions without human intervention [9]. In the last decade, the (IoT) has rapidly continued to penetrate more areas, including smart homes, smart cities, smart farming, Intelligent transportation systems, etc., [10].

According to reports, the scale of IoT is expanding exponentially, according to some statistics by Fan et al. [11] estimated that there may be 24.6 billion IoT connections by the end of 2025, with an annual growth rate of nearly 13%. Meanwhile, IoT connections, as reported by the International Data Corporation (IDC), are expected to exceed 38 billion by the end of 2025 and could reach over 50 billion by 2030 [12]. According to Saqib et al. [13], according to reports cited by Gul et al., 41 billion IoT devices may be present by the end of 2025, which might result in the generation of 175 zettabytes of data. Gul et al. [14] stated that there are about 50 billion device connections, and by the end of 2025, there are projected to be 75.44 billion device connections [15].

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The analysis of these enormous amounts of data is consolidated into decisions on many scenarios and management applications. Addressing their integrity and security issues is undoubtedly necessary because these data are crucial in decision-making. Otherwise, IoT applications will be severely hampered. For example, in two typical IoT domains -Smart Home and Smart Health- protecting or compromising data in these domains is crucial [16]. However, IoT devices have constrained memory to store all the data they generate, necessitating the integration of cloud computing to address their data storage issues [17].

The IoT faces serious security challenges because traditional protection measures, such as firewalls and anti-malware software, are unavailable [18]. This is mainly because IoT devices are designed with limited memory and low energy consumption, making it challenging to implement such protection [19]. As a result, IoT devices are more vulnerable to attacks, botnets, and network interruptions. Attackers may extract sensitive data or control networks when IoT devices are deployed. Advanced technologies, such as machine learning (ML), blockchain, edge computing and quantum computing, have also increased the potential for attacks [20]. In addition, the rapid growth of connected devices has intensified security concerns in IoT, making it a crucial field of research.

Therefore, the widespread adoption of IoT requires robust security measures, such as authentication, authorization, privacy, access control, system setup, verification, and data management [21]. Sensitive information can be damaged or modified at any stage. The security of these devices and the privacy of the data they store, and process are not always guaranteed. For IoT to gain greater acceptance, it is essential to ensure the confidentiality of user data [6]. IoT may potentially impact and penetrate more domains [10].

However, people's inherent concerns about security and privacy must be addressed effectively [22]. IoT's heterogeneous and dynamic nature also brings another challenge to the solutions proposed for the IoT setup [23]. Assessing IoT security is challenging because users and devices need to validate applications and networks. Security assessments for IoT authentication are even more complex, and many authentications are required for users to manage access to all applications and devices on the network [24]. Malicious nodes can access the network by exploiting authentication loopholes.

IoT security solutions, including authentication and encryption, are evaluated based on software complexity and hardware requirements [24], [25]. Software complexity considers time and memory complexity, while hardware requirements measure area and power consumption [26]. However, asymmetric cryptographic algorithms, such as RSA, ECC, and Diffie-Hellman, offer robust security but are less efficient in resource-constrained IoT environments.

The motivation for this systematic literature review (SLR) is to provide a knowledge base for researchers and practitioners concerned with solving IoT authentication problems while incorporating compatible protocols and embedded technologies to understand IoT security problems better and develop better solutions. This work presents the early schemes developed and explored through decades of advancements.

Subsequently, the research identified some research questions (RQs) and objectives (Section I-A), which connect the key concepts with the theme. Some of the contributions of this research are as follows.

- 1) Analyzing the security problems of IoT schemes and assessing their authentication schemes.
- 2) Investigating the security issues of deployed IoT devices and associated threats and attacks.
- 3) Illustrate, compare, and contrast robust authentication models' strengths, weaknesses, and other attributes.
- 4) Identifying open questions and predefined objectives to develop a clear framework for addressing the challenges within the IoT domain.
- 5) Establishing a clear roadmap to explore IoT trends and guide future research to resolve IoT security problems.

This SLR comprehensively reviews current research, fundamental concepts, and future directions in IoT authentication models, protocols, and embedded technologies. It identifies challenges, achievements, gaps, and new approaches for security features in IoT environments, offering valuable resources for researchers and practitioners.

This article begins with the Introduction in Section I, highlighting the problem, motivation, contributions, RQs & objectives. Next, Section II reports the methodology, the search process, the screening strategy, and all the systematic review methods. Section III presents the state-of-the-art problem in IoT. The subsequent Section IV discusses enabling technologies in IoT. Section V covers the security requirements in IoT, while Section VI examines the cyber-attacks on IoT devices. Lightweight IoT protocols are discussed in Section VII. Section VIII highlights some distinctions between IoT and the conventional Internet, while Section IX presents the challenges, open issues and research gaps in IoT. Finally, this article concludes with challenges and more contributions in Section X.

A. Research Questions Formulation

The RQs are presented here to support the grounds for conducting this review. These questions could guide researchers and practitioners to identify research gaps and other open issues for further investigation and propose solutions:

RQ1: What are the IoT domain's predominant security and privacy concerns?

Objective: To investigate and analyse the IoT domain's notch security and privacy issues to identify key vulnerabilities and concerns.

RQ2: What security threats exist in the IoT paradigm, and what countermeasures have been developed to mitigate them?

Objective: To identify security threats in the IoT environment and comprehensively review available countermeasures and mitigation models.

RQ3: What are the most compelling authentication schemes and algorithms developed and implemented in the IoT environment?

Objective: To evaluate and compare the effectiveness of existing authentication schemes designed for IoT in terms of security, scalability, and usability.



Fig. 1. Methodology mapping steps.

RQ4: How do formal and informal security methods evaluate the authentication process in IoT?

Objective: To assess IoT authentication processes' formal and informal security evaluations, focusing on the methodologies and criteria used.

RQ5: What are the optimal addressing protocols for IoT for identifying and managing IoT devices?

Objective: To evaluate the suitability of addressing protocols for identifying and managing IoT devices, explicitly focusing on adopting and implementing lightweight protocols.

RQ6: What technologies can be integrated with IoT and be compatible to support mitigating the constraints in IoT?

Objective: To study and analyze the newest technologies that could be integrated with IoT and that could support their robustness in security.

II. METHODOLOGY

This SLR was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) standards [27]. The study followed the guidelines from Kitchenham and Charters (2007) [28]. To guarantee complete reporting, prior investigations were followed [6]. This SLR's mapping procedure involves six steps, as shown in Fig. 1. These steps include the preliminary study method (Section II-A), keywords search (Section II-B), searching process (Section II-C), screening strategy (Section II-D), inclusion/exclusion criteria (Section II-E), and eligibility scoring process (Section II-F).

A. Preliminary Study Method

A first investigation was conducted to gain a better insight into the topic. This step motivates authors to find relevant RQs, keywords, and study scopes. During the initial phase, three activities were completed: identifying keywords, designing the search process, and identifying the screening strategy. We describe in detail how we undertake this study, search, select, deselect, extract knowledge, present, depict figures, and so on in the following sections.

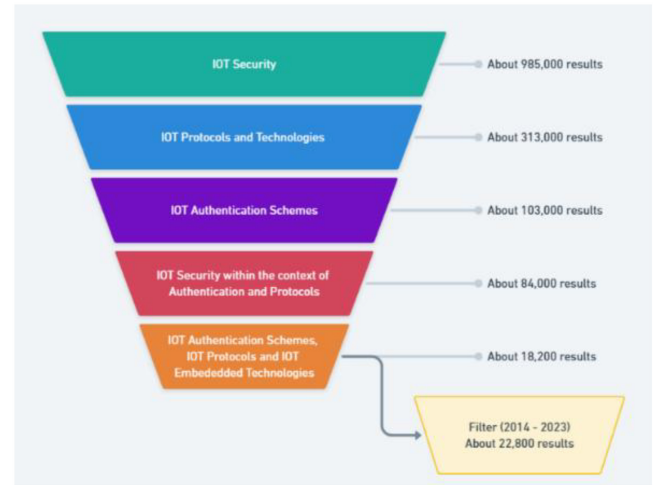


Fig. 2. Keywords search by google scholar.

A first investigation was conducted to gain better insight into the topic matter. This step was undertaken to motivate the identification of relevant RQs, keywords, and study scopes. During the initial phase, three activities were completed: identifying keywords, designing the search process, and developing a screening strategy. In this preliminary step, a dep search was used to conduct searches, select and deselect sources, extract knowledge, present findings, depict figures, and perform related tasks, as described in detail in the subsequent sections.

B. Keywords Search

To begin, a Google Scholar search was undertaken using the phrases "IoT security," "IoT Privacy," "IoT authentication schemes," "IoT protocols," "IoT embedded technologies," and "lightweight IoT security algorithms" to find relevant articles on the topic. The search was further refined by identifying relevant versions and new keywords. This stage involved screening relevant publications to address the IoT problem. The retrieved documents were used to identify relevant keywords and broaden the search scope. The keywords were used to search for more relevant papers on IoT authentication schemes, protocols and technologies.

Two authors of this article (Jameel Yalli and Mohd Hilmi) conducted this task. Fig. 2 shows the results of the keyword search and the number of publications found.

C. Searching Process

In this stage, we narrowed the search to a specific IoT security domain using applicable search parameters. We then searched using titles, abstracts, and keywords to identify relevant IoT papers. Journals, conference proceedings, and book chapters published between 2014 and 2023 were chosen to represent the past decade.

First, we explored the Open Athens website provided by the Universiti of Teknologi PETRONAS, which gives its staff and students access to some subscribed databases. We opted for the most reputable databases, Science Direct, Scopus, and

Web of Science. The keywords listed in Section II-B were searched.

The following describes the actions performed in each database.

- 1) *Science Direct (Elsevier)*: filtered results displayed 2572 articles without applying any filter. Then, some delimiters were eliminated, such as scope, type of article, magazine, topic, language, etc. The following is a summary of the results in the Science Direct database: number of articles per year. Year: 2014 (44 articles), Year: 2015 (40 articles), Year: 2016 (72 articles), Year: 2017 (98 articles), Year: 2018 (187 articles), Year: 2019 (277 articles), Year: 2020 (355 articles), Year: 2021 (449 articles), Year: 2022 (528 articles), Year: 2023 (472 articles). The type of article, the subject area and the access type are the same across the search criteria: the kind of article is 'research article', the subject area is 'computer science', engineering and social sciences, and the access type is 'open access' and 'open archives'. These articles include but are not limited to journals, such as IEEE IoT Journal, IoT, King Saud University—Computer and Information Sciences, Computer Networks, Network and Computer Application Magazines, Computer Science and Electrical Engineering, Future-Generation Computer Science, Computer Communications, etc.
- 2) *Scopus (RELX)*: We use the same keyword as in the Science Direct database to search for articles in Scopus. 'Range', 'type of article', and 'type of access' are all the same, but only the subject area and journal in which the article is published have some differences. This section presents works from the above-mentioned journals and the Advances in IEEE Access, IEEE IoT journals and Intelligent Systems & Computing. Journal articles 2014: 7, 2015: 11, 2016: 23, 2017: 27, 2018: 51, 2019: 61, 2020: 77, 2021: 89, 2022: 117, 2023: 185.
- 3) *Web of Science WoS (Clarivate)*: The keywords used to search for the article are the same as in the Science Direct database. A few filters are included here, including 'short filters' for 'open access', 'language' for 'English' and a few other selections. The yearly articles displayed are 2014: 15, 2015: 19, 2016: 39, 2017: 52, 2018: 109, 2019: 187, 2020: 108, 2021: 150, 2022: 154, 2023: 323. The menus/items listed here are those from which selections were made, but there are other menus from which no choice was made, so the search covers all of the items under that menu. For example, items such as 'computer science', 'telecommunications', 'communication', and 'engineering' categories were selected under the 'Research Areas' menu.

The distribution of articles across databases (Fig. 3) shows that ScienceDirect contributed the majority (59%, 2572 articles), followed by the Web of Science (26%, 1156 articles) and Scopus (15%, 650 articles). It is worth mentioning that Scopus has fewer articles due to topic titles and journals indexed, with each database having its delimiters and the ability to extract articles under specified items, resulting in a smaller number of articles.

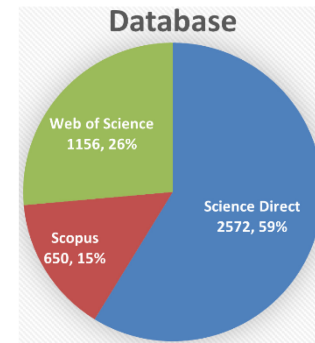


Fig. 3. Number of articles by database.

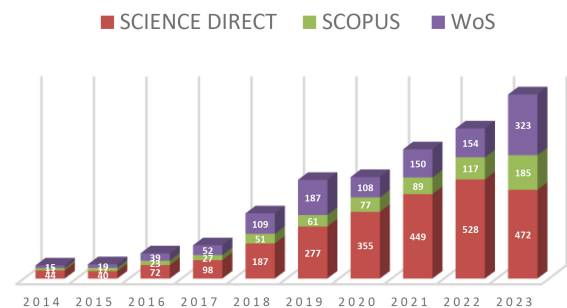


Fig. 4. Number of articles by year.

A total of 4378 articles were analyzed across three scientific databases, and relevant papers were identified based on the PRISMA Guidelines. Duplicate studies were removed using EndNote 20.5 and Microsoft Excel, and screening strategies were applied until 100 articles were included. The last search was conducted on 15 December 2023. The PRISMA-compliant flow diagram (2022) was followed [29], [30].

Fig. 4 Illustrates the number of publications obtained across the years from 2014 to 2023. A slight decline in publication numbers was observed in 2020, possibly due to the impact of COVID-19. The number of publications in 2022 is close to that of 2023; however, since the articles were compiled, screened, and synthesized during the last quarter of 2023, it is expected that more publications will appear online by the end of the year, leading to an increase in the final count. Three authors (Jameel Yalli, Umar and Mujeeb) conducted the search and the extraction of filtered articles.

D. Screening Strategy

After a database search using keywords and a broad search string, 3945 distinct publications were screened through three filtrations plus inclusion, exclusion, and criteria. Three authors (Jameel Yalli, Abdulrasheed, and Dahiru) evaluated the screening procedure and collaborated to finalize the results.

To confirm that the search results are appropriate to our SLR search keywords, they were joined using 'OR' and 'AND' logical operators and wildcards. The revised search was undertaken across three scientific databases. However, the original search results show a significant number of articles.

TABLE I
PAPER INCLUSION AND EXCLUSION CRITERIA

Criteria	Inclusion Criteria	Exclusion Criteria
Database	Science Direct, Scopus and Web of Science	Other databases
Publication year	2014 – 2023	Out of the range 2014 – 2023
Model stage	Implemented	Concept only
Relevance	IoT security and the RQs	Not relevant to IoT security and the RQs
Article type	Journal papers, Review papers, conference proceedings and book chapters	Technical reports, interviews, letters, tutorials, seminars, and/or blogs
Field	Computer science, engineering, networking and telecommunication	Other fields
Availability	Full-text articles	Abstracts only and in-press
Language	English	Non-English

E. Papers Inclusion and Exclusion Criteria

This section outlines the criteria used to include or exclude studies and is divided into two sections for a detailed explanation.

1) *Papers Inclusion Criteria*: To narrow down the search further, some criteria were applied to the articles' titles, abstracts and keywords, as listed in Table I.

2) *Papers Exclusion Criteria*: All publications were carefully screened using the inclusion and exclusion criteria indicated in Table I.

These criteria guided the selection of papers that directly addressed the RQs. Articles that do not match the required criteria were excluded. Works in which the proposed model is not implemented.

F. Eligibility Score and Quality Assessment (ES&QA)

In the first round of screening, a formal data analysis was conducted to assess the relevance of papers on IoT security and technologies. However, 1801 papers were retained, necessitating a second round of screening to ensure that all objectives were met. Advanced search features in Adobe Acrobat Reader and Foxit PDF Reader were utilised to identify specific articles related to authentication techniques, resulting in 1690 papers selected for further analysis.

After the second round of screening, 1590 papers that did not satisfy the eligibility criteria were removed to ensure their exclusion from the final list. The evaluation was based on specific criteria, including objectives, methodologies, limitations, and findings.

Fig. 5 and Table II describes the ES&QA. articles rated 8, 9, and 10 are acceptable, whereas those rated 1 are of the lowest quality, and those rated seven are not allowed. One hundred fifty-six eligible papers were ultimately included in the qualitative evaluation, and the publications that did not fit were eliminated. Two authors decided which papers were ineligible for inclusion (Low Jung and Safwan Al-Selwi).

Quality and Eligibility Score

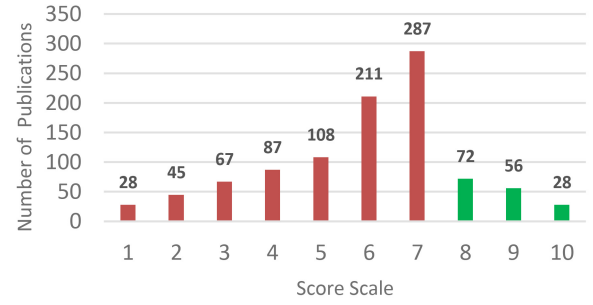


Fig. 5. Eligibility score scale.

TABLE II
CRITERIA FOR ELIGIBILITY SCORE AND QUALITY ASSESSMENT

CRITERIA	SCORE	DESCRIPTION
Unimplemented model	1	If the model is not implemented, the concept is explained.
Unstated limitations/scope	2	That is if the work is not confined to a particular problem or solving technique.
Undefined RQs	3	If the work does not formulate any RQs and tries to answer those questions.
Undefined objectives	4	If the work does not mention any objective, it tries to achieve succinctly.
Unclear methodology	5	That is if there is no transparent methodology for how the work is being carried out,
Unstated findings	6	That is if the work does not point out any novel findings to support its claims.
Unstated contributions	7	That is, if the work does not contribute to the body of knowledge.
Implemented / RQs / ROs	8	Any work implementing its model using experiments, simulations, or testbeds has stated RQs and ROs.
Implemented / RQs / ROs / Clear methods	9	That is any work that has implemented its model, stated its RQs & ROs and described its methodology.
Implemented / RQs / ROs / Clear methods / Contributory works	10	That is any work that has implemented its model, stated its RQs and ROs, described its methodology and proved its contributions.

Finally, Fig. 6 This summary summarizes the PRISMA-guided study selection process, which narrowed the 4378 initial records to 100 studies in the final review after applying screening and eligibility criteria.

III. STATE OF THE ART PROBLEMS IN IOT

The IoT ecosystem needed advanced security in both physical and virtual states [31], [32], but challenges like reusability, interoperability, and security [33] Are increasing due to the growing number of connected devices. Privacy and safety are crucial to ensure the full benefit of IoT applications, as widespread adoption may lead to massive information disclosure due to billions of devices storing data [33], [34], [35]. The main problem with IPv4 is its limitation, while the problems associated with IPv6 are security, routing, and mobility [36].

Since IoT infrastructure is similar to the backbone of the Internet because the Uninterrupted Data Protocol

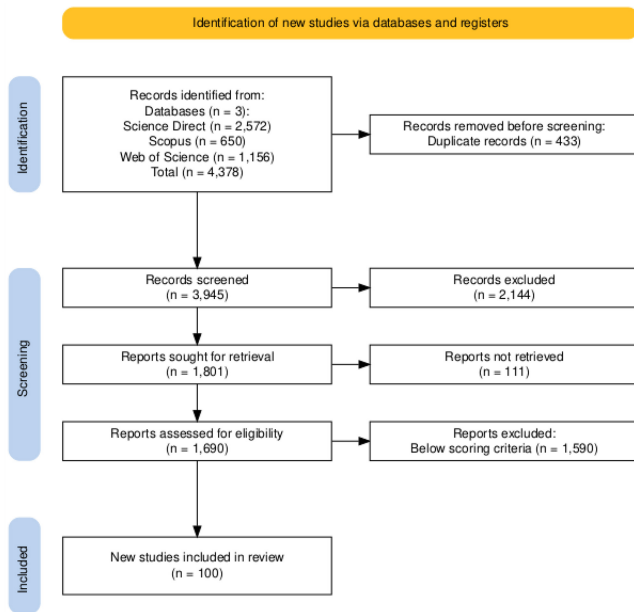


Fig. 6. PRISMA flowchart.

(UDP) is unstable, transmission data is performed using the Transmission Control Protocol (TCP). However, TCP also suffers from connection configuration, congestion control, and data buffers. In connectivity configuration, the resources used for IoT communication are mainly sensors, RFID tags, and PDAs. Still, they cannot handle all the configuration data needed because they do not have sufficient capacity and capabilities [6], [37].

The challenges of IoT security are rising every day and becoming more sophisticated. However, security problems related to connected new devices are increasing. Over the past 10 years, we have only been concerned about protecting our computers. Five years ago, we were worried about the safety of our smartphones. Today, the trend is to worry about safeguarding objects like cars, home appliances, wearable devices and many other IoT devices. Hewlett-Packard reported that 70% of the most common IoT devices were infected in 2014 [6], [37].

Chetan and Shahabdkar [38] examined several security issues related to heterogeneity, interoperability, resource constraints, privacy and scalability. Din et al. [42] explored security issues in IoT devices, highlighting their challenges due to their unique characteristics, such as low-cost design, device constraints, heterogeneity, privacy requirements, and trust management. Traditional security solutions, such as public-key encryption and IP-based solutions, are often inadequate, making them easier to compromise [39].

Among other numerous challenges of the IoT, there are software and network constraints [40]. Emerging technologies like blockchain, cloud computing, Intrusion Detection Systems (IDS), and ML can help provide security solutions and address associated threats and attacks in the IoT environment [41].

LightTrust is a lightweight approach for Industrial IoT nodes. It utilizes a centralised trust agent for secure data transmission and demonstrates resilience and robust environments through simulations [42]. Rai et al. [43] Explained that a

PUF-based authentication mechanism for IoT devices involves a key exchange between nodes and servers, ensuring secure communication between devices—the proposed scheme by Kumar et al. [44]. It uses lightweight RFID authentication protocols like LMAP, RAPP, and Kazahaya to construct long-range groups in decentralized and centralized data centre topologies for IoT devices.

LMAP reduces delays in IoT scenarios. However, scalability is limited due to network delays and lightweight protocols, potentially limiting its applicability in higher-security environments. The study by Porambage et al. [45] introduced PAuthKey, a lightweight authentication mechanism for secure communication in resource-constrained WSNs (WSNs) within distributed IoT applications. It outperforms conventional DTLS in low-resource devices. However, implicit access control and multicasting certificates could be extended to large-scale distributed IoT network applications.

Extensible Authentication Protocol—Trust Extension Protocol for Authentication of newly deployed Objects and Sensors through the Manufacturer (EAP-TEPANOM) is a lightweight authentication protocol optimized for IEEE 802.15.4 networks. It reduces packets, data usage, and header size, making it ideal for large-scale IoT deployments and enhancing EAP infrastructure efficiency [46]. However, the study should improve scalability and efficiency by integrating and optimizing TEPANOM with the EAP infrastructure.

The works by Fan et al. [11] improved a lightweight RFID mutual authentication protocol with a reader cache (LRMAPC) to reduce costs and security in low-cost RFID systems, with an enhanced version, ultraLRMAPC (ULRMAPC), offering stronger security and optimized performance. Although efficiency and security are increased by LRMAPC and ULRMAPC, their dependence on a reader cache raises the possibility of scalability and synchronization problems in large-scale or highly dynamic RFID applications.

Srinivas et al. [47] introduced a novel hashing-based authentication scheme for WSNs, ensuring high accuracy, user-friendly password changes, and robustness against man-in-the-middle and replay attacks. While Dhillon and Kalra [48] proposed a lightweight biometric remote user authentication and key agreement scheme for IoT services, validated through AVISPA to ensure robustness against multiple security attacks. Whereas Fan et al. [49] introduced the Ultra Lightweight Mutual Authentication Protocol (ULMAP), a lightweight, efficient, and secure protocol that uses Bit and XOR operations to prevent DoS and synchronization attacks. However, its reliance on light operations like XOR and hash functions may hinder its ability to withstand sophisticated attacks.

According to Shen et al. [50], a secure multilayer authentication protocol and session key generation method for Wireless Body Networks (WBANs), combining energy-efficient sensor node group authentication and ECC certificateless cryptography, as Gope et al. [51] introduced a lightweight, privacy-preserving authentication protocol for RFID systems using ideal PUFs, with an enhanced version for noisy environments, improving security and efficiency—however, Fan et al. [52] our approach is a lightweight authentication protocol that effectively combats security

threats and maximizes efficiency. It utilizes less than 0.5% of look-up tables (LUTs) and registers.

The works by Banerjee et al. [53] introduced a lightweight, efficient, and secure IoT anonymous user authentication protocol using hash functions, PUF, and XOR operations, backed by formal analysis, AVISPA verification, and NS3 simulation. While Ostad-Sharif et al. [54] proposed a secure and lightweight authentication and key agreement protocol for IoT-based WSNs, verified using AVISPA and demonstrated to outperform existing protocols. Both schemes' enhanced security and performance, but their reliance on AVISPA validation tools may overlook real-world deployment complexities and evolving attack vectors in dynamic IoT environments.

Khalid et al. [55] introduced a biometric-based authentication and key agreement protocol (BUAKA), a four-phase biometric authentication and key agreement protocol, which provides robust security against attacks while minimizing computational cost and communication overhead. While Ahmed et al. [56] introduced elliptic lightweight cryptography with authentication (ELCA), a secure IoT cybersecurity system that integrates authentication, encryption, and ECDH for key distribution, ensuring confidentiality and authenticity through mathematical security proofs. Whereas Chen et al. [57] proposed a firmware-secure multifactor authentication protocol (FSMFA) that uses PUF and firmware integrity to enhance IoT device and system security, ensure mutual authentication, and efficiently secure key exchange.

However, the models BUAKA, ELCA, and FSMFA show-case improvements in IoT security by integrating cutting-edge mechanisms like firmware integrity, biometric-based authentication, and cryptographic integration; they mainly depend on specific computing methods and favourable circumstances. This dependence might restrict their scalability and adaptability in various real-world IoT deployments, where resource limitations, heterogeneous devices, and changing attack vectors could present serious difficulties.

Siddiqui et al. [58] proposed a lightweight PUF—PKI Digital Certificate Authentication Scheme for IoT, enhancing security and efficiency by integrating PUFs with PKI, ensuring mutual authentication, and suitable for resource-constrained devices. Whereas Saqib et al. [13] This article proposes a secure IoT authentication scheme using passwords, identities, and digital signatures, utilizing ECC and hash chains. However, due to its computational complexity, this scheme may pose challenges for resource-constrained devices. Relying on computationally intensive cryptographic operations may not be practical for resource-constrained devices and does not address scalability challenges [59].

The work by Pu et al. [60] introduced a lightweight, privacy-preserving mutual authentication and key agreement protocol (PMAP) for the Internet of Drones (IoD) environment. It uses physical, unclonable functions and chaotic systems to secure communication between drones and zone service providers (ZSPs). The scheme is implemented using the Henon chaotic map for randomness and tested on a real-world testbed. However, it has a vicious circle in key updates, high redundancy and potential vulnerabilities. Therefore, Xiang and

Technique

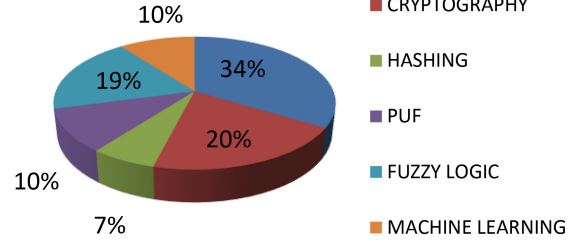


Fig. 7. Techniques used in IoT security.

TABLE III
COMPARISON OF DIFFERENT AUTHENTICATION SCHEMES

Ref.	AUTHENTICATION TYPE	ALGORITHM DEVELOPED	SECURITY MECHANISM	VULNERABILITY
[44]	Lightweight protocol	Light weight auth protocol (LMAP), RFID Auth Protocol with Permutation (RAP) & kazahaya	Multiple DC interconnected	Availability is not guaranteed
[45]	Lightweight authentication and keying mechanism	PAuthKey	Implicit certificates and end-to-end security	Group Implicit certificates for key management are not utilized in large-scale WSN
[46]	Lightweight and secured protocol	EAP-TEPANOM Authentication protocol	ECC certificates & WAP-TLS employed	Architecture needs to be interrogated
[11]	Lightweight and Ultra-lightweight authentication	lightweight RFID mutual authentication protocol with cache in the reader (LRMAPC)	RFID tag utilized	Tracking, spoofing, and replay attacks are quickly launched into the protocol.
[48]	Lightweight biometric user authentication	It uses hash operations and XOR operations	Authentication phases with password	No testbed is defined, and lightweight features are not extended to IoT devices.
[49]	Ultra lightweight mutual authentication	Ultra Lightweight Mutual Authentication Protocol ULMAP	Pseudonyms & XOR operations employed on NFC technology	Suffering attacks, including clone attacks, MITM attacks, and packet loss attack
[50]	Lightweight multi-layer authentication	PDA authentication protocol	Non-pairing certificateless anonymous authentication	The ECC operation and exponential key size become inefficient and vulnerable
[51]	Lightweight anonymous authentication	Generation algorithm FE.Gen and Reconstruction algorithm FE.Rec	RFID and PUF were used to identify and verify nodes	The scheme is vulnerable to side-channel attack, physical attack
[53]	lightweight anonymous user authentication for the IoT environment by	Novel, lightweight, physically secure, anonymous algorithm	One-way cryptographic hash function, PUF and bitwise exclusive-OR (XOR) operations are utilized	The scheme becomes vulnerable to some attacks
[54]	Lightweight authentication key agreement	Lightweight authenticated key agreement protocol	High-Level Protocol Specification Language (HPLSL) utilizes four back-ends and abstraction methods.	Exposes the secret long-term keys of parties, no perfect forward secrecy
[62]	ECC-based authentication & key agreement model	RSEAP: RFID-based Secure and Efficient Authentication Protocol for Vehicular Cloud Computing	Verified using the AVISPA simulation tool and formal methods, it proves security against replay attacks and MITM attacks.	Potential computational overhead, as compared to simpler protocols, relies on substantial cryptographic operations.
[63]	Enhanced hash-based protocol with ECC (Elliptic Curve Cryptography)	A Secure Lightweight Authentication Scheme for Next-Generation IoT	Secure against attacks, such as impersonation, replay, DoS, and MITM attacks, it shows formal and informal security analysis	Susceptible to computation overhead and storage costs due to increased cryptographic operations in resource-constraint IoT devices
[13]	A Lightweight Three-Factor Authentication for Critical IoT Framework	ECC-based scheme with hash chains	Efficient mutual authorization of entities; resistant to MITM, replay, and impersonation; verified using Scyther tool.	Computationally complex in IoT environments due to elliptic curve operations and hash chains
[58]	PUF—PKI-based digital certificate authentication technique with certificate authorities	An Improved Lightweight PUF—PKI Digital Certificate Auth Scheme for the IoT	Resistant to many conventional attacks and mutual authentication issues	Computational complexity in generating keys relies on the strength of PUF functions and PKI infrastructure
[55]	lightweight and less complex authentication	Biometric-based User Authentication and Key Agreement Protocol (BUAKA)	One-way hash functions and fuzzy extractors for sensor nodes are utilized	Sensor nodes become vulnerable to spoofing, sinkhole attacks, Sybil attacks, DoS attacks and eavesdropping
[56]	Lightweight cryptography authentication	Encryption Lightweight Cryptography and Authentication (ELCA)	Cryptography and authentication are merged utilizing the elliptic curve Diffie-Hellman (ECDH)	The scheme is vulnerable to specific session attacks
[57]	Multi-factor authentication	Firmware-Secure Multi-Factor Authentication (FSMFA)	PUF is employed to provide security to the hardware, while device firmware is to protect the software gateway to the outside world. [69].	Firmware is tampered with by implanting malicious programs vulnerable to cloning, modification, and attacks.

Liu [61]. It proposes a more straightforward, more effective mechanism for maintaining efficiency and security in the IoD environment.

Finally, Fig. 7 It illustrates the distribution of techniques used to solve problems in IoT security. Authentication is the most commonly used technique, accounting for 34%, followed by cryptography at 20% and fuzzy logic at 19%. Techniques like PUF (10%), ML (10%), and hashing (7%) are also employed, albeit to a lesser extent [32].

Table III compares different authentication schemes with integrated technologies, describing its security mechanism and vulnerability.

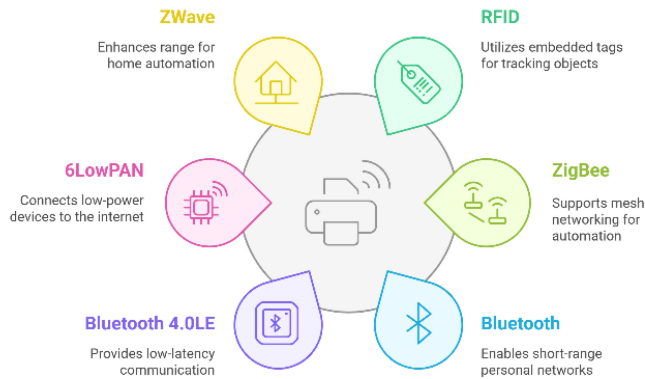


Fig. 8. Enabling technologies in IoT.

IV. ENABLING TECHNOLOGIES IN IOT

Radiofrequency identification (RFID) is the first technology for communicating objects. The RFID system works to authenticate three main substantial components: Tags, readers, and data centers [64]. The readers can scan the tags before obtaining the required information and then save it in the data center.

Wireless technology has evolved from RFID technology, where the chip does not have to have a reader visible before communicating with it. Unlike the barcode technology that requires the physical visibility of the reader to communicate [65]. RFID is an actuator that can trigger events that are not possible with a barcode.

WSN is a multilevel wireless network connected to a distributed sensor array that measures a particular data collection device's speed, humidity and temperature and transmits the values to the processing device. RFID is a short-range asymmetric communication technology, while WSN technology enables long-range peer-to-peer communication.

As IoT aims to enable automatic data connections and data sharing between entities and any device capable of detecting, analyzing, transmitting, and storing information over the Internet, the network has become heterogeneous due to the diverse backgrounds of objects. Early technologies include RFID, WSN, Bluetooth, WLAN, wireless urban network (WMAN), cellular network (LTE, 2G, 3G, 4G, 5G, now 6G) [66], [67], [68].

In recent years, WSN and RFID have aroused great interest in IoT applications. Both technologies have different capabilities and are used in scenarios based on their requirements. RFID enables the reading of content to identify and recognize the objects associated with the tag. WSN creates dynamic materials depending on the environment used. These technologies are used as cheap, short-term technologies. Bluetooth creates personal networks that communicate with or without a line of sight. Bluetooth is associated with IoT due to its low power consumption, but its ability to communicate directly with the Internet is limited. It usually involves intermediate nodes, such as PCs, as a gateway to the outside world [69].

Fig. 8 illustrates these enabling technologies and the following is a list of their description.

- 1) *Radio Frequency (RF) or RFID*: is the first technology to connect a communication device, with communication

ranges ranging from 100 m to 1 kilometre depending on the transmission power and the antenna [65]. RFID uses small embedded electronic chips called tags to identify and track objects. RFID was initially recognized as an efficient, low-cost mobile communications technology between IoT devices but was later discovered to be limited in communication directly or indirectly to the Internet because it could increase device size, cover longer distances, and be only close to the Internet [70], [71].

- 2) *ZigBee*: ZigBee Alliance developed a low-power wireless protocol for short-range networks used in lighting systems, home automation, smart energy management, and industrial controls. It supports mesh networking, low energy consumption, and scalability [72].
- 3) *Bluetooth*: is a low-cost, short-range technology that is part of the IEEE 802.15.1 family. Bluetooth connects with or without physical sight and creates a personal network. Although Bluetooth's low power consumption makes it suitable for the IoT, its ability to communicate directly with the Internet is limited. Typically, it has an intermediary node that serves as a gateway to the outside world, such as a PC [69].
- 4) *Bluetooth 4.0LE*: is a subset of Bluetooth 4.0, the old name of which was WiBree. It features a new protocol stack and profile architecture whereby it enables connectivity and connectionless MAC for low latency and fast communication [73].
- 5) *6LowPAN*: is an expanded or more recent form of IPV6. This technology supports IPV6 networks and is a low-power wireless LAN. IPV6 provides enough address space to recognize and categorize anything on Earth. Traditional Web servers, like HTTP and TCP/IP, use the same IP-based network standard protocols as sensor nodes. Protocols for addressing are IPV6 and 6LowPAN [74].
- 6) *ZWave*: ZenSys Technologies develops a low-power wireless protocol for automation in residential and commercial environments. It operates on a mesh network architecture, improving range and reliability in smart home applications [72].
- 7) *WiFi*: Wireless Fidelity is a family of the most commonly used IEEE 802.11x and is the most widely used method for connecting devices to the Internet for both IoT and conventional use [75]. All our devices, such as laptops, have a WiFi interface that communicates with wireless routers and then provides both ways of access to the Internet. The WiFi standard family allows the construction of a short-distance wireless network. Types of WiFi networks include IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, IEEE 802.11e for Quality of Service (QoS) extension, IEEE 802.11f for transfer management and IEEE 802.11i for security extension [76], [77].

Other advanced technologies include symmetric key encryption operations that do not use many resources, as they are faster than bitwise functions, XOR, and permutation operations and are well suited to IoT. A significant difference between

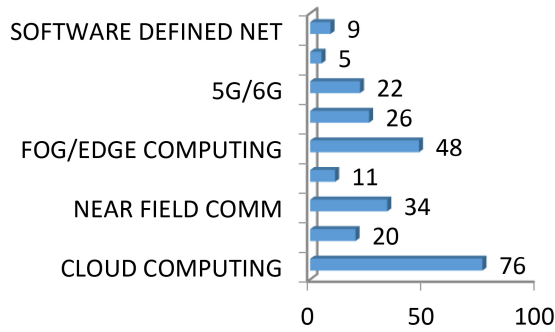


Fig. 9. IoT security in different domains.

symmetric algorithms is between stream encryption, hashing function, and block encryption. AI and ML are considered promising security solutions for IoT, and the domain is used to design models based on empirical datasets and predict future trends [78].

Algorithms, such as support vector machines (SVMs), K-means clusters, naive Bayes, artificial neural networks (ANNs), ambiguous logic, genetic algorithms and automated stack encoders, are used to develop IoT security models [76]. Blockchain is another widely accepted area that can be used to record computations between nodes and users in the IoT network and to record financial transactions in cryptocurrencies [79]. Blockchain comprises four technologies: consensus algorithms, encryption, distributed ledgers, and smart contracts. The main security features of blockchains used by IoT include decentralization, transparency, mutability, nonrepudiation, pseudonymity, and tracking. These technologies and their properties make them suitable for IoT networks. Blockchains can encrypt data at rest, store data, and digitally sign it [80].

Table IV shows multiple technologies integrated with IoT to proffer optimum solutions to its security problem. While Fig. 9 shows the number of publications found in the respective technologies integrated with IoT.

V. SECURITY REQUIREMENTS IN IoT

This section presents a comprehensive literature review illustrating the protocol stack developed by IEEE and the IETF research group. Key IoT security requirements include authentication, confidentiality, integrity, availability, resilience, nonrepudiation, privacy, liability, anonymity and trust [87]. This study examined the security protocols, encryption mechanisms, key management techniques and end-to-end authentication of IoT environments and measures to mitigate threats, such as denial of service (DoS), ultrawideband direct sequence (UWDS) and chip propagation spectrum (CPS) [88].

Through this review, researchers identified several challenges in the IoT environment, including but not limited to standards, mobility support, traffic characteristics, QoS, authentication, data integrity, and privacy concerns [89]. Providing security across IoT systems requires addressing these challenges through security, integrity, authentication, authorization, availability, and privacy [90].

The following key aspects are essential for addressing IoT security challenges.

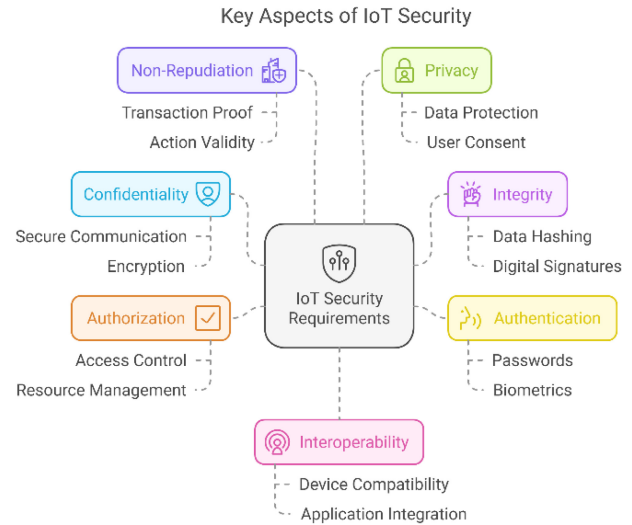


Fig. 10. Taxonomy of security requirements.

- 1) **Confidentiality:** In IoT, confidentiality can be defined as protecting sensitive data from unauthorized access by other nodes or users. It is to ensure that only authorized access by authorized users is allowed. Data viewed during transmission or stored at transient in IoT systems compromises confidentiality [91]. Therefore, privacy protection starts with data safety and the prevention of data breaches. Security measures, such as secure communication protocols, encryption, access control, etc., maintain confidentiality in IoT environments [92].
- 2) **Integrity:** Integrity in IoT security ensures that data and information remain accurate and unchanged throughout its life cycle within IoT systems. It protects against unauthorized modifications, manipulations, or data corruption and ensures IoT-generated information's reliability. Measurements, such as data hashing and digital signature, help to maintain the truthfulness of data in the IoT [92], [93].
- 3) **Authentication:** The IoT security authentication verifies the identity of devices and users seeking to access the IoT system [94]. It ensures that only authorized entities can interact with IoT devices within the network. Standard authentication methods include passwords, biometrics, digital certificates, and token-based systems, which improve the security and credibility of the IoT ecosystem [95].
- 4) **Authorization:** IoT security defines the actions and resources authenticated users and devices can access or control within the IoT system. It identifies permissions and restrictions to prevent unauthorized acts and ensures the appropriate use and protection of IoT resources and data [65].
- 5) **Nonrepudiation:** An action or transaction performed in an IoT security cannot be denied by the device and does not guarantee the authenticity or validity of actions or transactions performed in the IoT system. It proves that specific actions and communications occurred and were not compromised and contributes to building trust and responsibility in IoT interactions [96].

TABLE IV
DIFFERENT TECHNOLOGIES USED IN OTHER DOMAINS, OBJECTIVES, PROS, AND CONS

Ref.	DOMAIN	TECHNOLOGY	OBJECTIVE	ADVANTAGES	LIMITATIONS
[44]	Datacenter	RFID	To provide lightweight authentication built on simpler verification protocols	The scheme merges multiple authentication models and executes well	Availability is guaranteed
[45]	WSN	6LowPAN, CoaP, RPL, IEEE, 802.15.4	To provide certificate CA by cluster head CH	Cluster head CH issues implicit certificates CA	It has little support to access data sensing network entities
[46]	WSN	WLAN	To create a protocol required to integrate closely the TEPANOM solution and its architecture with the EAP infrastructure	This EAP supports various authentication procedures by introducing less communication overhead than others. It does not require specific global infrastructure, so it is scalable.	It could not integrate the TEPANOM solution, and it could not incorporate its architecture with EAP.
[11]	WSN	RFID	To provide ultra-lightweight authentication by utilizing the RFID cache reader	It attains lesser computational cost precisely when verifying a large number of tags.	It requires expanding memory space for a considerable number of tags.
[48]	WSN	GWN SN	To create a lightweight biometric technique based on remote user authentication and key agreement for secure access	Users can be authenticated either remotely or offline	Memory requirement needs to be out in the testbed, and the lightweight feature can be extended to real IoT devices
[49]	NFC	RFID	Creating a mutual authentication protocol with ultralight weight achieves forward security with subkeys and sub-index in the key update.	From an economic point of view, this approach is light, practical, and straightforward in defending against synchronization attacks.	NFC authentication must improve its functions and performance while taking into account system security and privacy
[81]	WSN	WLAN	To design a light Weight device-to-device continuous authentication protocol	It authenticates data transmitted between two devices within a distinct period in the IoT ecosystem	It is difficult to find more dynamic device features and to develop more accurate models to minimize battery energy consumption
[82]	Cloud computing	WLAN	To design a lightweight multifactor secure smart card on a remote user authentication scheme	It is an efficient, lightweight multilayer authentication mechanism with ECC encryption and safe session group encryption methods.	ECC operation adds to the computation cost as well as memory space
[83]	SN	PUF	The goal is to design a mutual authentication technique between edge devices belonging to CE-based IoT	The mutual authentication scheme claims to use strong PUFs, which enable it to provide a reliable response when a challenge is sent.	The attack surface of terminal nodes is extended because many nodes are deployed in an unsupervised environment, which is more likely to be attacked physically.
[84]	SDN	MQTT	To develop a lightweight authentication protocol leveraging the security features of MQTT	Data transmission between IoT entities is efficiently done using the lightweight Message Queue Telemetry Transport (MQTT) protocol.	Only one protocol is utilized, neglecting other necessary and supporting IoT protocols.
[85]	Blockchain	Blockchain	To develop a new lightweight authentication protocol-based blockchain	The blockchain features of immutability and decentralization of control are utilized	The actual implementation needs to be tested
[13]	IoT Security	ECC and Hash Chains	To develop a secure, efficient, and lightweight three-factor auth model for IoT applications with password, identity, and digital signatures.	It ensures mutual auth between the subscriber, broker, and publisher. It is resistant to MITM, replay, session key, and impersonation attacks and validated with the Scyther tool.	There is computational overhead due to ECC operations and challenges for highly resource-constrained IoT devices.
[58]	IoT Security	PUF & PKI (Public Key Infrastructure)	To design a lightweight PUF-PKI-based digital certificate auth technique for IoT with certificate authorities to enhance security	It provides robustness and resistance against replay, impersonation, session hijacking, and message tampering, and it has been validated using Tamarin Prover.	The strong cryptographic primitives of key management using PUF & PKI bring more computational complexity.
[60]	Internet of Drones (IoD)	PUF chaotic systems and session key protocols	To propose a lightweight and privacy-preserving mutual auth and key agreement protocol (PMAP) for securing communication between drones	As validated via AVISPA and formal security analysis, it resists replay, impersonation, message modification, drone capture, MITM, and known session key attacks.	Computational complexity due to using PUFs and chaotic systems; potential energy consumption constraints for highly resource-limited drones.
[55]	IIOT	WSN	To create a lightweight biometric-based user authentication and key agreement scheme that is computationally cheap for IoT	The protocol is lightweight, resulting in less complex authentication. It assures the authenticity, integrity, and availability of data packets.	The protocol needs further investigation.
[86]	IBA	MQTT	Propose an authentication architecture from device to device with interaction as a security factor for identity.	The simulation and informal analysis can support IBA dynamic adaptability, scalability, and resistance to several known attacks.	The scheme needs further investigation.

6) *Privacy*: IoT security privacy protects individual personal data and information IoT devices collect. It includes the implementation of protections and strategies to control the collection, access and use of data, the respect of user

privacy rights and the prohibition of misuse or disclosure of personal data without consent [88], [97].

7) *Interoperability*: It ensures that devices in the IoT systems communicate seamlessly while maintaining

security standards. It enables a coherent and safe IoT platform where different devices and applications can interact seamlessly and effectively without compromising security measures [98].

- 8) *Security and Privacy*: This is at the forefront of IoT security. Security measures defend IoT techniques from unauthorized access and manipulation, while privacy protection ensures that personal data is collected, processed, and shared in accordance with user regulations and consent. The two are essential to building trust in the IoT [99].
- 9) *Discovering New IoT Applications*: This has necessitated the need for creative thinking and security considerations. The creativity leads to an investigation of new use cases while ensuring that security controls are included from the beginning to safeguard against attacks and likely vulnerabilities [100]. Other issues include a lack of tools for managing massive datasets, third-party data, the growing threat of public databases, data leakage, and concerns about digital security. The notion of intelligent homes and cities is based on embedded systems, intelligent technologies, and IoT. In general, smart cities are based on information technologies and connected infrastructure to enable greater living standards and decision-making [92], [101], [102].

VI. CYBER SECURITY ATTACKS IN IOT

Attacks on routing protocols cause the network to collapse. Due to limited computational capacity, processing power, and limited storage, traditional routing protocols are not suitable to be implemented on IoT and WSN Ad Hoc networks [103]. IoT is composed of heterogeneous networks, including Ad-hoc networks, WSNs, the Internet, 3G networks, 4G networks and now 5G [104]. DDoS is the most common attack in IoT due to the heterogeneity and complex nature of its networks, which exposes transport layers to attacks and the solution is to update the system and implement a DDoS detection framework [105], [106]. A single technology is insufficient to achieve the best security for IoT; more than one technology is needed.

Several launched attacks were threatening the security of IoT extracted by this work from the studied literature, which could pose a threat to IoT and its network [107]. To achieve the optimum security for IoT, it is imperative to integrate more than one technology into a single model. For example, a weak application layer will compromise the network security regardless of how much data privacy effort is put into the perception layer [108]. Thus, corresponding technologies for cross-layer can be designed [34].

To design the technology for the cross-layer approach and have complete security, all the attacks and threats listed in

Table V needs to be addressed. In addition to addressing heterogeneous integration issues, information transmission security, assurance of CIA (confidentiality, Integrity & availability) responsibility, facing data leakage, data incompleteness, etc, to build a new system autonomy and

heterogeneous integration model to meet the requirements of different layers to obtain different data through different layers of IoT networks [109].

This work has studied fifty attacks commonly launched on IoT networks. The models proposed in the literature try to counter some of these attacks and discover new trends in some scenarios. For simplicity, these 50 attacks are presented in Table V listed some attacks found to be sent against IoT networks with a brief description, their potential threat, and a provisional countermeasure.

VII. LIGHTWEIGHT IOT PROTOCOLS

The Internet Engineering Task Force (IETF) has developed IoT-specific security solutions [90]. Likewise, the IEEE has established standards relating to IoT, such as the 802.11 x wireless network protocol. We found that most researchers preferred MQTT as the primary communication protocol in their projects because of its high network capacity, large network capacity, smaller package sizes, and more efficiency than alternative communication protocols.

Several commonly used IoT precise application protocols, such as Constraint Application Protocol (CoAP) and Message Queue Telemetry Transport (MQTT), do not provide adequate security protection. Therefore, they rely on the basic security protocols at the transport layer, which are computationally intensive. Protocols for messaging and electronic platforms are vital for communication between IoT devices. This work has investigated the most IoT messaging protocols and compared their efficiencies. These protocols include MQTT, CoAP, AMQP, and DDS.

MQTT is faster than other protocols due to the vast network bandwidth and small packet transfer [13]. MQTT is one of the dependable protocols for achieving reliable data exchange. It depends on a central broker located in the cloud, which often has problems with network congestion, performance overhead, or bottlenecks. Therefore, fog computing is introduced to develop distributed architectures containing multiple breaks. MQTT is a publish/subscribe messaging protocol that enables many clients (publishers and subscribers) to communicate with one another via a central broker. It is one of the light, flexible, and simple IoT application layer protocols, with a few required messages to be exchanged per connection, small headers, and optional service options, such as the control QoS level and the implementation of safe connections [110]. These characteristics make MQTT appropriate for resource-limited devices, low-bandwidth networks, and high-latency networks [111].

Fig. 11 illustrates an overview of IoT Communication Protocols and a short description of each protocol is as follows.

- 1) *Message Queue Telemetry Transport (MQTT)*: is a lightweight, publish-subscribe, node-to-node and device-to-device network protocol designed for message queueing services. It specializes in connecting devices to objects or entities to devices in a remote location that involves resource constraints entities and

TABLE V
IoT ATTACKS

NO	ATTACK	DESCRIPTION	THREAT	COUNTER MEASURE
1	Brute force attack	An attacker tries many passwords until he gets the correct one	Attackers could systematically guess the password	Employ biometric authentication
2	Code injection attack	Malicious programs are injected into a node or device in the IoT	Attackers could control certain functions	Authenticate the identity of the node
3	Cryptanalysis attack	Encryption keys maliciously inferred in the IoT systems	Data could leak, and key functions could fail	Encryption algorithm can be executed
4	Dos / DDoS attack	The network is flooded with substantial traffic, keeping available resources continuously busy	Network paralyzes	Increase network protection system
5	Deception attack	A malicious node pretends to be an ordinary node to trick access	In the IoT system, specific functions fail	Multi-factor authentication can be added
6	De-synchronization attack	It uses a control flag or a fake sequence to copy messages between endpoints of active connection	Attackers can desynchronization endpoints, so sensor nodes transmit the message again	Prevent the server in the one-way and two-way communication request
7	Eavesdropping attack	Theft of data or information over a wireless connection	Information outflow	A secret key can be set to filter noise data
8	Error detection attack	Preventing reliable delivery of digital data over unreliable channels	Information loss	Employing error detection correctly and monitoring
9	False data injection attack	False data is transferred into IoT applications	Incorrect response instruction is executed	False or error data is detected
10	Forward secrecy	Carrying key secrecy forward even if long-term secrets are compromised	Key leakage leads to information tampering	Strong encryption to prevent key leakage
11	Gateway forgery attack	Masking a GWN to become malicious and sending an attack	Malicious GWN penetrates any part of the network	Strong authentication of the GWNs
12	GWN bypassing attack	A GWN accessing the network without going through the authentication process	GWN could launch any attack on the network	Strong authentication of the GWNs
13	GWN impersonation attack	GWN poses as a trusted entity to gain access to sensitive information and resources	GWN could jeopardize sensitive resources at its disposal	Strong authentication of the GWNs
14	Insider attack	A legitimate node or user becomes malicious for some personal gain	The whole network is at risk	Continuous authentication among the nodes
15	Leak of verifier attack	The attacker steals sensitive data from the server and uses it to learn more about the network	Impersonate itself as a legitimate user	Strong authentication protocol
16	MAC Spoofing attack	Assigning a legitimate MAC address to a malicious node to access the network	The network always considers that particular MAC as a genuine	Employ a mechanism to verify all MAC addresses
17	Malicious script attack	During installation, confidential user data is stolen in the software	Confidential data leak	Dynamic monitoring mechanism can be added
18	Malicious virus attack	Attacker disguises and sends a self-propagating virus	System will crash	A reliable firewall can be deployed
19	Man-in-the-middle attack	The attacker scrutinizes the network flow and controls communication info between two common nodes	Data integrity and accuracy are lost to the attacker	A secure communication protocol can be deployed
20	Many logged-in users attack	A single ID is used to log into the system by several attackers	Attackers flood the network to launch many attacks	Single ID sign-in at a time
21	Modification attack	An attacker modifies the message sent from the sender to the receiver	Data integrity is compromised	Deploying the Dolev Yao model to protect the data
22	Mutual authentication	Two or more nodes in a network could authenticate one another	If one authenticates and the other cannot, it is malicious.	Prevent loopholes in the authentication algorithm
23	Node capture attack	Attacker tampers or even replace nodes in the network	Important information leakage	Examine and detect malicious nodes
24	Offline identity guessing attack	Trying to get the correct identity while in offline mode	Delay in detecting any attack since it is offline	Ensure registration nonces are not reusable
25	Offline password-guessing attack	Trying to guess the correct password while the system is in offline mode	If guessed right, full network privilege could be accessed	Strong authentication mechanism in offline mode

(Continued)

limited network bandwidth. MQTT runs over TCP/IP (TCP/Internet protocol) [112].

- 2) *Constrained Application Protocol (CoAP)*: is an IoT protocol for transferring packets to the Web. Unlike

MQTT, CoAP uses the Interruption Data Protocol (UDP); although they have some similarities, CoAP is standardized more than MQTT [113]. CoAP has a powerful ability to prevent network failures and use fewer

TABLE V
IoT ATTACKS

26	Parallel session attack	Two protocols are executed so messages in one are formed in the other	System may crash	Design one algorithm for one task only
27	Password change attack	Attackers take advantage of a user changing password to read and manage the process	Password cloning or recharge	Strong authentication to verify user changing password
28	Phishing attack	Attackers pretend to be a phishing website to trick user information	User account password is revealed	Be alert when users go online
29	Privileged insider attack	Attackers with admin rights in the system can make sensitive changes	Hard to find and fish out	Install new configuration
30	Repetitive attacks	Attackers gain the trust of the IoT through multiple scam inputs	The authenticity of the certificate is undermined	Threshold time can be set and repeat
31	Replay attack	The attacker captures the message sent and replays it to the sender or receiver	Pass the authentication phase of the system	Deploy a strong authentication in place
32	Resilience against node capture attack	Seizing a node physically and extracting data from its memory	Confidentiality is breached	Random key redistribution
33	Routing information attack	Information is spread and controlled by manipulating routing protocols	Data is lost, and some functions fail	A secure routing protocol can be deployed
34	Sensor captures attack	Attackers hijack the sensor to become under his control	Information leakage	Deploy sensitive algorithm
35	Sensor Node impersonation attack	The attacker stands between two nodes, A and B, that are not in direct range	Information leakage and packet dropping	Phishing detection algorithms
36	Session key computation attack	The session key could be stolen or monitored	Total information is at risk	Strong cryptography to protect cryptosystems
37	Session key disclosure attack	Attackers can steal the session key and launch a side-channel attack	The network could be destroyed	Strong cryptography to protect cryptosystems
38	Side channel attack	Leaked information is used to launch attacks on physical systems	Physical and software components could be destroyed	Authentication and strong cryptography
39	Sinkhole attack	The infected node becomes a circular forwarding node	Destruction to transmitted data, data is disclosed	Combine many security protocols
40	Sleep deprivation attack	Keeping the node active for a long time until it is constrained to shut down	Some nodes fail, and the system is low	Add & devise node energy storage capacity & technique.
41	Snooping attack	Attackers listen to the traffic on the network	All unencrypted data could be tapped and read by the attacker	Enable firewall to filter out any suspicious user
42	Stolen smart card attack	Smart cards are stolen, accessed and modified data on their chip	Personal information leakage	Embed unclonable PUF chip
43	Stolen verifier attack	Stealing user ID and password from the password table	Unauthorized access breached confidentiality	A reliable algorithm to prevent accessing the routing table
44	Sybil attack	A node behaves as multiple nodes with multiple identities	Data may leak and certain functions may fail in the network	Security identification mechanisms can be increased
45	Unauthorized attack	Information is obtained by accessing RFID short of authentication	Information leakage	RFID authorization access mechanism can be defined
46	User anonymity attack	The identity of the node in the transaction is not known	Any action could not be traced to its originator	Apply non-repudiation
47	User forgery attack	The attacker forged user's identity and performed actions to harm the network	An action could have severe consequences	Authentication and continuous authentication
48	User impersonation attack	Attacker masks to access and claim all rights in the network	The whole network is at risk	Authentication and continuous authentication
49	User tracking attack	Preventing tracking the correct nodes	Misleading the correct track information	Inspection and monitoring of the nodes
50	Wormhole attack	Send malicious packets over two malicious nodes	Causes network congestion and breakdown	Amend routing protocol

network resources than many others. CoAP provides an interactive request and response model supporting integrated services and resources [114]. CoAP meets

the requirements of the Web, such as multicasting, low overhead, and simplicity. In addition, a demand response model is provided for low-power networks. The CoAP

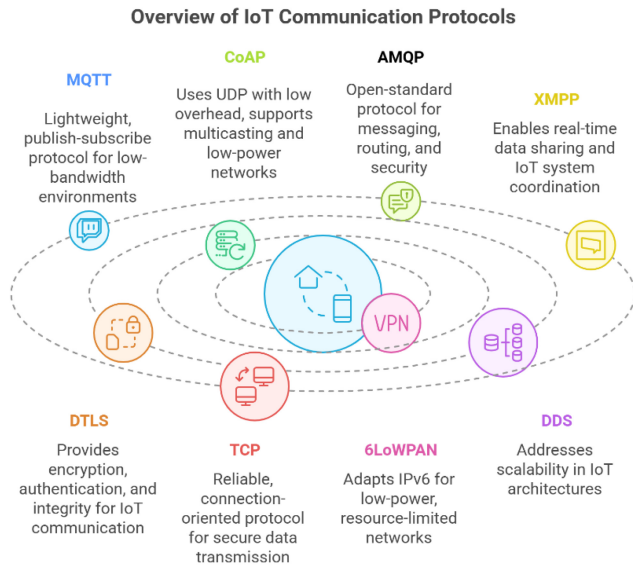


Fig. 11. Overview of IoT communication protocols.

protocol is mainly used for Web transport. RFC 7252 defines a limited application protocol as a specialized Internet application protocol for limited devices [115].

- 3) *Advanced Message Queuing Protocol (AMQP)*: is considered to be an open standard application layer protocol [116]. AMQP has features appropriate for IoT, including routing, queueing, messaging, orientation, and security. AMQP is also an open-source standard for asynchronous communication on networks. The application uses AMQP to communicate in an interoperable and encrypted way. The protocol is used for client/server messaging and IoT device management [117]. AMQP is available as a message-oriented intermediate in application layer protocols. Like MQTT, it uses TCP, but it generates a large number of messages and affects the amount of transmitted data [121, 122].
- 4) *Extensible Messaging and Presence Protocol (XMPP)*: is a real-time and versatile communication protocol suitable for IoT. It allows real-time message sharing and any other information from the devices and applications. It enables IoT devices to communicate, share, store, and transmit data among its entities, making it a helpful tool for managing and coordinating IoT systems and services. Examples of communication that is used in applications is audio and video calls [120].
- 5) *Datagram Transport Layer Security (DTLS)*: is a protocol that authenticates entities on an IoT network, especially if the network is untrustworthy. Like CoAP, DTLS uses UDP. It is used to encrypt, authenticate, secure data, increase security, and demonstrate the integrity of IoT networks in terms of device connections, low latency, and real-time data transfer [121].
- 6) *TCP* is a reliable and connection-oriented communication protocol used on the Internet and in IoT device networks. This protocol facilitates orderly and secure data processing. It guarantees data supply, data transfer by order, and error control, making it more effective

and complex than UDP in applications that require data integrity and reliability [89], [90].

- 7) *IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN)*: It is similar to IPV6 and allows the identification and labelling of IoT devices. 6LoWPAN uses Internet Control Communication Protocol (ICMP) to control access, identify inaccessible destinations, and find neighboring nodes. 6LoWPAN allows efficient IPv6 package transmission over resource-limited wireless networks. By optimizing packet size and address, you reduce overhead and energy [122]. Middleware creates standard application requirements and links applications from many domains to various interfaces [101].
- 8) *Data Distribution Service (DDS)*, like CoaP and DTLS, is utilized to address scalability issues in IoT architecture, which are not addressed in the IoT architecture.

VIII. DISTINCTIONS BETWEEN IOT AND INTERNET

IoT can be seen as an extension or advancement of the Internet. However, each stands to be a separate network with its technologies, although its operation mode is nearly the same. Table VI shows some of the significant distinctions between IoT and the Internet [10]. In this work, we have shown thirty differences between the Internet and the IoT.

IX. CHALLENGES, OPEN ISSUES AND RESEARCH GAPS

IoT faces many challenges, affecting its development, and its platform, operating system, communication channel and devices are exposed to a wide range of security threats [31]. IoT systems are vulnerable to attacks that use devices, such as attack channels, physical manipulation and communications violations. Security technologies are essential to protect IoT entities by combating network layer attacks, communication encryption, sleep deprivation, and resource exhaustion attacks. Despite the progress made since its invention in 1999, IoT still faces a persistent security problem [123]. Fig. 12 and the following list contains possible open issues in IoT and offers researchers and stakeholders the opportunity to design innovative systems that enhance security and address these vulnerabilities effectively [124].

- 1) *Scalability*: It refers to the ability of an IoT network to adapt to the growing number of devices and data without significant performance degradation or major system changes. Scalable IoT infrastructures are crucial to managing the rapid expansion of IoT applications and ensuring long-term functionality [125]. As the adoption of IoT continues to grow exponentially among individuals and organizations, it is essential to address scaling challenges to prepare for the future of IoT technologies.
- 2) *Heterogeneity*: IoT heterogeneity refers to the diversity of devices, protocols, and data formats in ecosystems due to different manufacturers and communication standards [126]. Effective management of this diversity is crucial to ensure seamless data exchange, scalability, and interoperability between devices. IoT platforms and middleware have played an essential role in removing these

TABLE VI
DIFFERENCES BETWEEN IOT AND THE INTERNET

#	Criteria	IoT	Internet
1	Devices	Uses RFID nodes & WSN nodes.	Uses PCs, servers, and other general-purpose devices.
2	Algorithms	Primarily utilizes lightweight algorithms due to resource constraints.	It can use a combination of lightweight and complex algorithms.
3	Connection Type	Often it uses slower, less secure wireless connections.	Typically, use faster, more secure wired or wireless connections.
4	Node Security	Nodes are more easily compromised due to limited resources and security mechanisms.	Devices are more complex to compromise and require skilled attackers.
5	Societal Acceptance	Slow acceptance as its new smart applications evolves.	Widely accepted and integrated into society.
6	Vulnerability	More vulnerable to attacks due to weak security and distributed nature.	Vulnerable too but generally more secure than IoT.
7	Protocols	Uses lightweight protocols (e.g., CoAP, MQTT, 6LoWPAN).	Supports a wide range of standard protocols.
8	Information Requirement	Often requires personal information for application functionality.	Many applications can operate without requiring personal information.
9	Operating System	Uses specific operating systems like TinyOS or RIOT OS.	Uses generic operating systems like Windows, Linux, and macOS.
10	Resources	Limited resources (e.g., power, memory, processing) and weaker network protection.	More abundant resources and robust network protection.
11	Impact	Focuses on convenience and automation.	Focus on improving productivity, communication, and information sharing.
12	Data Challenges	Directly leads to significant data challenges due to massive data generation.	It can result in significant data challenges as data volumes grow over time.
13	Deployment Environment	Devices are often deployed in hazardous or outdoor environments.	Gadgets are typically deployed in controlled environments like homes, offices, and schools.
14	Security Treatment	Security issues are treated as standalone components and are not unified.	Security is more integrated and addressed holistically in architecture.
15	Architecture	Customized, with no single framework universally applied.	Standardized architecture, such as the OSI model, is widely used.
16	Network Type	Relies on low-power and lossy networks (LLNs).	Typically uses dynamic topologies for scalability and performance.
17	Data Loss	More prone to significant data loss due to frequent attacks and constrained resources.	Data loss is minimized but can still occur due to large-scale failures or attacks.
18	Cryptographic Technology	Supports lightweight cryptographic algorithms designed for constrained devices.	Supports complex cryptographic algorithms for stronger security.
19	Layer Stack	Uses a simplified stack (perception, network, and application layers).	Follows the traditional 7-layer OSI model.
20	Perception/Datalink Layer	Perception layer nodes have limited power and computational ability.	The data link layer can accommodate higher-capacity devices.
21	Network Layer Vulnerability	Vulnerable to routing table modifications and other attacks.	Vulnerable to man-in-the-middle and counterfeit attacks.
22	Application Layer	Encounters issues like data privacy, access control, and information disclosure.	Primarily focused on managing applications, data, and services.
23	Physical Layer Protocol	Often uses 6LoWPAN over IPv6.	It often uses WiFi as the physical layer protocol.
24	Network Layer Protocol	Uses DTLS for secure communication.	Uses TCP for secure and reliable communication.
25	Application Layer Protocol	Uses CoAP for lightweight communication.	Uses HTTP/FTP/SMTP for robust communication.
26	Security Architecture	It aims to integrate seamless communication among devices, objects, and people.	Primarily designed for user interaction and less for machine-to-machine communication.
27	Data Transfer Protocols	It uses CoAP for transferring bits, bytes, and packets.	It uses HTTP, FTP, or SMTP to transfer packets and frames.
28	Data Exchange Protocol	Uses MQTT for constrained device communication.	Uses TCP/UDP for general-purpose data transmission.
29	IP Addressing	Uses IPv6/6LoWPAN for efficient IP assignment to constraint devices.	Uses IPv4 and IPv6 for IP assignment.
30	Connection Standards	Uses IEEE 802.15.4e for low-power IoT connections.	Uses IEEE 802.3 (Ethernet), 802.11 (WiFi), and related families for broader connectivity.

differences, facilitating data integration, and developing and maintaining robust IoT solutions.

- 3) *Integration and Coordination*: They refer to seamless combinations of various devices, systems and data sources to ensure their efficiency and reliability [127]. This ensures smooth information exchange and collaboration between IoT components and

improves automation, data analysis, and decision-making. Effective integration and coordination are crucial to unlocking the IoT's full potential and enabling the development of more sophisticated and interconnected applications and services [128].

- 4) *Standardization*: The lack of standardization in IoT is due to the lack of universally accepted protocols and

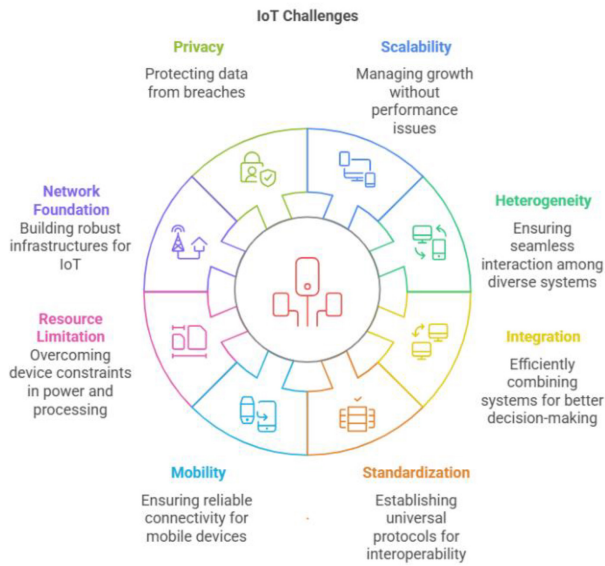


Fig. 12. Key challenges in IoT.

guidelines for devices and ecosystems [126]. This fragmentation leads to interoperability problems, security vulnerabilities, and difficulties in naturally integrating IoT nodes and platforms. While ongoing research aims to establish common standards to address these issues, a coherent and secure IoT landscape requires a united framework.

- 5) *Mobility*: IoT mobility refers to the integration of IoT chips into mobile objects, such as humans, vehicles and wearables, so that these devices can move while remaining connected [126], [129]. For example, wearable devices and vehicle sensors. To ensure the reliability of data connectivity and transmission for mobile IoT devices, real-time tracking, tracking, monitoring and management are essential components of effective IoT implementation [129].
- 6) *Resource Limitation*: The IoT's resource limitation stems from the inherent limitations of its devices, including processing capabilities, power consumption, memory and bandwidth [123]. IoT devices often operate with minimum battery life, computing capacity, and communication resources, presenting challenges for widespread deployment. Effective resource management and optimization are essential to ensure IoT devices and systems' efficient and sustainable operation.
- 7) *Network Foundation*: IoT base includes communication infrastructures connecting devices, including Wi-Fi, mobile networks and WLAN, WMAN, WBAN [127]. A robust network is vital for efficient and reliable data transmission, enabling IoT devices to operate effectively and provide valuable insights. Network technology selection depends on factors adapted to the specific needs of IoT applications, such as energy consumption, range, and data volume.
- 8) *Privacy*: It is an essential challenge for IoT, which has hampered its widespread acceptance [127]. Although efforts are being made to integrate security measures

TABLE VII
PRISMA 2020 CHECKLIST

HEADING	ITEM	SECTION
Title	1	Title
Abstract	2	Abstract
Introduction	3	Section 1
	4	Section 1.1
Methods	5	Section 2.4.1, 2.4.2
	6	Section 2.3
	7	Section 2.4
	8	Section 2.5
	9 - 15	Not applicable
Results	16a	Fig. 5
	16b	Not applicable
	17	Section 3
	18 - 22	Not applicable
Other Information	24	Not applicable
	25	Acknowledgement
	26	Declaration
	27	Not applicable

in the design and implementation phases, more work is needed to address privacy issues effectively. During attacks, proactive measures must be taken immediately to protect personal and location information. Lack of privacy protection and standard operating procedures (SOPs) remains a significant obstacle to adopting IoT worldwide.

- 9) *Trust*: It is essential to ensure secure and reliable communication between devices [125]. Nodes must provide security levels to ensure trust in the data they manage and establish safe relationships between devices and entities within the IoT network. This ensures reliable communication, prevents unauthorized access, and protects against data breaches, making trust the cornerstone of the IoT system's integrity and security.
- 10) *Ownership*: The ownership of IoT refers to the trust and control that users and organizations have over the devices and the data they generate [130]. Establishing trust ensures the accuracy of data, the safety and reliability of devices, and the application of privacy standards. Ownership includes legal and operational control over IoT nodes and their data, which is essential to establishing user confidence, promoting the adoption of IoT, and preserving the power and privacy of connected assets and information.

X. CONCLUSION AND CONTRIBUTIONS

This work gives a summary of the SLR conducted in the Section X-A and highlights some more contributions to the body of knowledge in the Section X-B.

A. Summary

This SLR explores the critical challenges, security requirements and fundamental aspects of IoT systems and highlights key areas, such as scalability, heterogeneity, mobility, resource constraints, network foundations, privacy, trust and ownership. The rapid adoption and integration of IoT in various fields will

TABLE VIII
ABBREVIATIONS

Acronym	Meaning
6LowPAN	IPv6 Over Low Power Area Network
AES	Advanced Encryption Standard
AMQP	Advanced Message Queuing Protocol
ANN	Artificial Neural Network
BUAKA	Biometric-based User Authentication and Key Agreement Protocol
CoaP	Constraint Application Protocol
CPS	Chip Propagation Spectrum
DDoS	Distributed Denial of Service
DDS	Data Distribution Service
DES	Data Encryption Standard
DoS	Denial of Service
DTLS	Datagram Transport Layer Security
EAP-TEPANOM	Extensible Authentication Protocol - Trust Extension Protocol for Authentication of New deployed Objects and sensors through the Manufacturer
ECC	Elliptic Curve Cryptography
ECDH	Elliptic Curve Diffie-Hellman
ELCA	Encryption Lightweight Cryptography and Authentication
ES&QA	Eligibility Score and Quality Assessment
FSMFA	Firmware-Secure Multi-Factor Authentication protocol
GN	Gateway Node
HF	high frequency
HLSL	High-Level Protocol Specification Language
HTTP	Hyper Text Transfer Protocol
IDS	International Data Corporation
IEEE	Institute of Electrical and Electronic Engineering
IETF	Internet Engineering Task Force
IoT	Internet of Things
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IT	Internet Technology
LF	Low Frequency
LMAP	Lightweight authentication protocol
LTE	Long Term Evolution
MQTT	Message Queue Telemetry Transport
NFC	Near Field Communication
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PUF	Physical Unclonable Function
QoS	Quality of Service
RAPP	RFID Authentication Protocol
RFID	Radio Frequency Identification
ROR	Random Or Real
SLR	Systematic Literature Review
SN	Sensor Node
SVM	Support Vector Machine
TCP	Transmission Control Protocol
UDP	Uninterrupted Data Protocol
UHF	Ultra-High Frequency
ULMAP	Ultra Lightweight Mutual Authentication
UWDS	Ultra-Wideband Direct Sequence
WBAN	Wireless Body Area Network
WLAN	Wireless Local Area Network
WMAN	Wireless Metropolitan Area Networks
WPAN	Wireless Personal Area Network
WSN	Wireless Sensor Networks
XMPP	Extensible Messaging and Presence Protocol

provide enormous opportunities, but it will also face significant vulnerabilities and operational challenges. To address these problems, a robust security mechanism, effective resource optimization and standardized protocols are required to ensure that IoT systems are secure, efficient and reliable and meet the growing demands of users and organizations.

The findings highlight the need for innovative solutions to manage the inherent complexity of IoT, including seamless integration, data interoperability and protection against evolving threats. Increasing user trust through ownership, privacy protection, and reliable infrastructure is essential to the widespread acceptance of IoT. By addressing these open issues, researchers and stakeholders can contribute to developing robust and scalable IoT solutions that unlock their full potential while preserving user confidence and security.

B. Additional Contributions to This Survey to the Body of Knowledge

In this research, many works on the security of IoT authentication and the state of the art lightweight algorithms within the IoT ecosystem were studied, and the work has contributed the following:

- 1) Analyzing 100 journal articles over the past decade, focusing on changes, improvements, and progress in proposed solutions to IoT security issues, as illustrated in Fig. 4.
- 2) This work presents various technologies researchers use to address IoT security issues, including Blockchain, Cryptography, Cloud & Edge Computing, 5G, and IDS. Key technologies like protocols, enabling technologies, architecture, authentication, encryption, and key management are also considered. This provides insights into standard technology usage and helps new researchers choose suitable technologies.
- 3) The analysis shows that authentication, encryption, key management, advanced protocols, and security architecture are commonly used to solve IoT security problems, accounting for 34% of research.
- 4) This work aims to benefit practitioners and researchers by providing a comprehensive view of security problems and solutions in IoT over the past decade, thereby expanding knowledge in this field.

APPENDIX

The PRISMA 2020 Checklist was acquired from https://www.prisma-statement.org/s/PRISMA_2020_checklist-arf9.docx (last viewed Jan 25, 2025). Table VII shows the checklist used in this SLR, including the part, number, and placement in the publication.

ABBREVIATION

Table VIII provides the abbreviations used in this article.

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